Lab 1 - Product Description

Julian Diaz

CS 411 W

Professor Hosni

2/7/26

1.0.1

Table of Contents

1 Introduction.....3

2 Product Description.....4

2.1 Key Product Features and Capabilities.....4

2.2 Major Components (Hardware/Software).....5

Figure 1: Major Functional Component Diagram (MFCD).....5

3 Identification of Case Study.....6

4 Glossary.....7

5 References.....10

List of Figures

Figure 1: Major Functional Component Diagram (MFCD).....5

1 Introduction

Managing large amounts of data on a daily basis, increases the risk of mistakes and errors. Dealing with invoices manually follows this same principle. Mistakes made with manual invoices consist of data field entry errors or delays in presenting deliverables in a timely manner. Therefore, the repetitive nature of this workflow and its risks opens an opportunity for a solution, and the flexibility for staff to redirect their attention to more significant tasks.

Under a magnifying glass, ADS receives approximately 150,000 invoices a year, from about 3,000 suppliers (ADS Pitch Document 2025). The AP team is only six people, and they must comb through around 50 invoices per person, daily. This is not ideal for anyone, invoices pile up, the likelihood of errors increase, the staff is distressed, and the suppliers are unsatisfied. Difficulties with invoice processing is not universal, however organizations like ADS handling so many invoices at a time face complications in their workforce.

AI^2 (Artifically Intelligent Invoices) is a web-based intelligent automation system designed to streamline invoice management and matching. The objective of AI^2 is to reduce the manual workload, increase turnaround time of invoices, and deliver data of the highest quality possible. This approach enables the AP team to spend their time effectively and efficiently on more unique discrepancies. The inclusion of the AP team also refines the effectiveness of AI^2 with some of its functional components.

2 Product Description

AI² is responsible for automating invoice intake, and matching it to a corresponding purchase order. Alongside that, it's capable of receiving feedback from human reviewers to improve its matching algorithm. The goal of AI² is to reduce the amount of manual labor and processing an individual has to face, and leave the repetitive work to automation. Therefore, consumers that use AI² can invest their attention to special cases where invoices and purchase orders don't match up.

2.1 Key Product Features and Capabilities

By integrating with Outlook GCC High using Microsoft Graph API, the automated invoice intake feature would automatically retrieve and parse invoice attachments and through an email's body text. Instead of manually obtaining the invoice and parsing the information within, AI² will automatically process it.

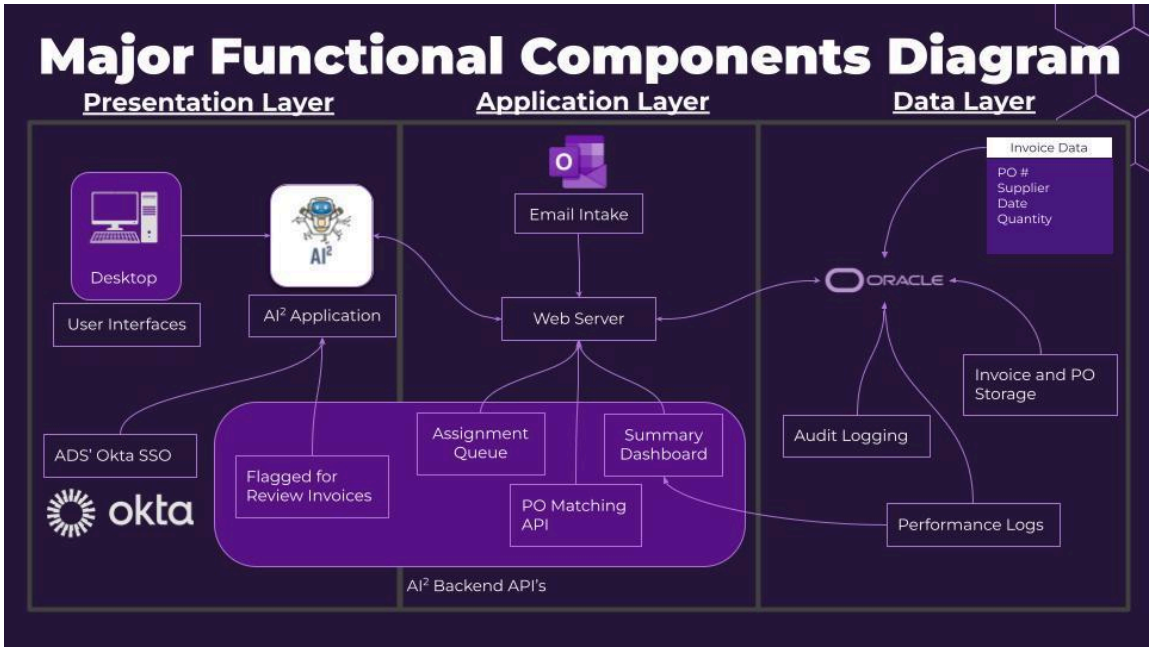
Connecting to Oracle NetSuit through Alteryx API for data retrieval, matching invoices to purchase orders, line items, and amounts. Smart matching continues AI²'s objective, reduction of repetitive tasks and endless searching to allow a productive workflow. A feature running after smart matching has concluded is confidence scoring. Lower confidence scores on a match are flagged for human review. This feature identifies discrepancies between what was interpreted and what matched, and gives the ability to be refined based on what was modified.

Queue assignment is automated to increase productivity, but reassignment is also available through an administrator. And lastly, audit logging is present for compliance and traceability.

2.2 Major Components (Hardware/Software)

AI² is hosted in AWS GovCloud for security and government compliance, with access to the application through standard office workstations and laptops. The software architecture is constructed in a web interface with Flask and SQLAlchemy for the database. Key integrations of AI² include Outlook for invoice capture and Oracle NetSuite for data exchange and auditability. Document and invoice scanning come from a multitude of different components to consider the different file formats invoices are received such as, AWS Textract and Aperture.

Figure 1: Major Functional Component Diagram (MFCD).



3 Identification of Case Study

ADS (Atlantic Data Systems) is a government contracting firm managing procurement and supplier payments. Currently they're facing issues with a heavy AP workload due to manual invoice processing, which leads to delays in supplier payments. At Team Copper, we plan on developing AI² by automating the repetitive processes in intake, sorting, and matching. Alongside giving diagnostic feedback and using human feedback to improve accuracy during matching. Similar challenges exist among other mid-to-large government contractors, logistics companies, and manufacturers. AI²'s modular architecture allows adaptation to other organizations with similar high-volume AP processes.

4 Glossary

AI² (Artificially Intelligent Invoices): The proposed automated invoice processing system designed to streamline Accounts Payable workflows through AI-driven matching, classification, and routing.

Alteryx: A data automation and analytics tool that enables secure API connections and data workflows between applications.

AP (Accounts Payable): The amount a business owes for goods and services purchased on credit; typically due at intervals of 30, 45, 60, or 90 days depending on repayment terms.

APEX: A strongly typed, object-oriented programming language that Salesforce developers use to execute flow and transaction control statements on the Salesforce platform.

AWS GovCloud: A secure U.S. government-compliant cloud environment that supports sensitive data and regulated workloads in accordance with FedRAMP and DoD requirements.

Atlantic Diving Supply (ADS): An American federal contractor company that provides equipment and logistics solutions to the Department of Defense, federal agencies, and first responders.

CMMC (Cybersecurity Maturity Model Certification): A framework developed by the U.S. Department of Defense to assess and enhance the cybersecurity posture of contractors handling federal information.

ERP (Enterprise Resource Planning System): An integrated software system used to manage core business processes, such as finance, procurement, and supply chain operations.

FedRAMP (Federal Risk and Authorization Management Program): A U.S. government program that standardizes security assessment, authorization, and continuous monitoring for cloud services.

Graph API: A Microsoft API interface used to access and automate data operations within Outlook and other Microsoft 365 services.

Invoice: A document listing goods or services provided, including a statement of the amount due; a bill.

Oracle: The system of record used by ADS for enterprise data and financial management.

PO (Purchase Order): The official order list ADS sends to suppliers detailing the items or services requested.

Salesforce: A robust Customer Relationship Management (CRM) platform that allows businesses to manage customer relationships efficiently.

Supplier: A person or organization that provides goods or services needed by another organization.

Vendor: A person or company offering goods or services for sale.

5 References

ADS Pitch Document Sample

Tamaro, R. (2025, March 4). *The Hidden Costs of Manual Accounts Payable and How Automation Solves Them - CASO Document Management.* CASO Document Management. https://caso.com/2025/03/the-hidden-costs-of-manual-accounts-payable-and-how-automation-solves-them/

AI Invoice Processing Benchmarks 2025 - Accuracy, Speed, And Cost Comparison. (2025, September 9). Parseur. Retrieved September 27, 2025 from https://parseur.com/blog/ai-invoice-processing-benchmarks#:~:text=Speed%20(according%20to%20SuperAGI),automation%20drives%20significant%20productivity%20gains

Jex, S. (2024, September 24). *Accounts Payable Automation by the Numbers: 10 Statistics to Know.* Quadient. https://www.quadient.com/en/blog/ap-automation-by-the-numbers-10-statistics-to-know

Avalanche of E-mail Coming Out of the Computer, Overwhelming the Person.(n.d.) 123RF. Retrieved September 29, 2025, from https://www.123rf.com/photo_176621030_avalanche-of-e-mail-coming-out-of-the-computer-overwhelming-a-person-eps-8-vector-illustration.html

Gartner Survey Shows That a Third of Accountants Make Several Financial Errors Per Week Due to Capacity Constraints. (2024, February 21). Gartner. https://www.gartner.com/en/newsroom/press-releases/2024-02-21-gartner-survey-shows-that-a-third-of-accountants-make-several-error-per-weeo-due-to-capacity-constraints

Accounts Payable Automation Trends: Key Insights and Latest Statistics. (n.d.) Kefron. Retrieved October 6, 2025, from https://kefron.com/2025/02/accounts-payable-automation-trends-key-insights-and-latest-statistics/?switch=us